

Qualification Pack



Industrial Electrician

QP Code: PSS/Q1706

Version: 1.0

NSQF Level: 4

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PSS/Q1706: Industrial Electrician

Brief Job Description

The Industrial Electrician performs pipe and plate earthing and industrial wiring as per planned specifications. The individual also performs installation and testing (upto 450V) of switchgears and cables, control panel wiring of industrial AC motors, testing and maintenance of transformer, operation of DG (Diesel Generator) sets and its controls as per specifications.

Personal Attributes

The individual must physically and mentally be able to perform essential functions of the job safely. The job holder must have an eye for detail, patience and discipline to carry out detailed and repetitive tasks. The candidate should be able to read, listen carefully and understand instructions and warnings.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [PSS/N1331: Apply basic health and safety practices for power related work](#)
2. [PSS/N1336: Working effectively with others](#)
3. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)
4. [PSS/N1717: Prepare testing and maintenance of transformer](#)
5. [PSS/N1716: Prepare control panel wiring of industrial AC](#)
6. [PSS/N1715: Prepare industrial wiring as per planned](#)
7. [PSS/N1714: Preparation installation and testing of Switchgears](#)
8. [PSS/N1713: Preparation pipe and plate earthing](#)
9. [PSS/N1712: Preparation for the operation of DG \(Diesel Generator\) sets and its controls](#)

Qualification Pack (QP) Parameters

Sector	Power
Sub-Sector	Generation, Transmission, Distribution



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Occupation	Operation and Maintenance - Generation, Erection, Installation and Commissioning - Generation, Erection, Installation and Commissioning - Distribution
Country	India
NSQF Level	4
Credits	18
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7411.0100, 7411.0301
Minimum Educational Qualification & Experience	12th grade Pass (with Science Background) OR Completed 2nd year of the 3-year diploma after 10 (In Relevant Field) OR 10th grade pass (with two year of any combination of NAC/NTC/CITS or equivalent in relevant trade) OR Diploma (Pursuing 2nd year of 3 year regular Diploma in relevant field after 10th) OR 11th grade pass with 1 Year of experience Relevant Experience OR 10th grade pass with 2 Years of experience Relevant Experience OR Previous relevant Qualification of NSQF Level (3.5) with 1-2 Years of experience Relevant Experience OR Previous relevant Qualification of NSQF Level (3) with 3 Years of experience Relevant Experience
Minimum Level of Education for Training in School	8th Standard
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	31/08/2026
NSQF Approval Date	31/08/2023
Version	1.0
Reference code on NQR	QG-04-PW-00870-2023-V1-PSSC



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NQR Version

1.0



Qualification Pack

PSS/N1331: Apply basic health and safety practices for power related work

Description

This unit covers health, safety and security for power related work. This includes procedures and practices that candidates need to follow to help maintain a healthy, safe and secure work environment in a power plant, power station/substation or on the field while working on power equipment. It covers responsibilities towards self, others, organizational assets and the environment.

Scope

The scope covers the following :

- Follow workplace health and safety practices
- Follow fire safety practices
- Follow emergencies, rescue and first-aid procedures

Elements and Performance Criteria

Follow workplace health and safety practices

To be competent, the user/individual on the job must be able to:

- PC1.** use protective clothing/equipment for specific tasks and work conditions (Protective clothing: leather or asbestos gloves, flame proof aprons, flame proof overalls buttoned to neck, cuff less (without folds) trousers, reinforced footwear, helmets/hard hats, cap and shoulder covers, ear defenders/plugs, safety boots, knee pads, particle masks, glasses/goggles/visors) (Equipment: hand and face shields, machine guards, residual current devices, shields, dust sheets, respirator)
- PC2.** identify job-site hazards and possible causes of risks/accident at the workplace (Hazards: electrical hazards, sharp edged and heavy tools, heated metals, oxyfuel and gas cylinders, welding radiation, hazardous surfaces, hazardous substances, physical hazards. Possible causes of risk and accident: physical actions, not following instructions, inattention, sickness and incapacity, health, not taking safety precautions)
- PC3.** use standard safe working practices when working at heights, confined areas, trenches and with electrical equipment
- PC4.** follow warning signs and symbols while working with electrical systems (Warning signs: Danger plate, high voltage area, out of service, etc.)
- PC5.** ensure positive isolation of electrical equipment and system as per the given standards
- PC6.** identify any abnormalities in electrical equipment or system viz temperature, pressure, flow of current etc.
- PC7.** use various methods of accident prevention in the work environment (Methods of accident prevention: use of equipment and working practices, safety notices, advice, instruction from colleagues and supervisors)
- PC8.** use proper scaffolds, elevated platforms and ladders for working at height
- PC9.** inspect various tools and plants (T & P) routinely for any signs of oil, water leakages
- PC10.** store flammable materials and machine lubricating oil safely and correctly



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PC11. check that the emission and pollution control devices are working properly in line with environmental policy standards

PC12. inform the relevant authorities about any abnormal situation/behavior of any equipment/system without delay

Follow fire safety practices

To be competent, the user/individual on the job must be able to:

PC13. use various appropriate fire extinguishers for different types of fires correctly (Types of fires: Class A, B, C, D and E)

PC14. use appropriate rescue techniques during fire hazard

PC15. use good housekeeping practices at all times to avoid accidental fire

Follow emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

PC16. administer appropriate first aid to the victims for the relevant injuries (Injuries: Bleeding, burns, choking, electric shock, poisoning etc.)

PC17. act promptly and appropriately to an accident situation and administer basic first aid including CPR to the victims of cardiac arrest due to electric shock or medical emergency in real

PC18. write an incident report and send the same to the authorized person

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. names (and job titles if applicable) and location of all documents along with the people responsible for health and safety in a workplace

KU2. possible causes of risk and accident: physical actions, not following instructions, inattention, sickness and incapacity (such as drunkenness), health hazards (such as untreated injuries and contagious illness), not taking safety precautions

KU3. safe working practices when working with tools, machines

KU4. positive isolation of electrical equipment and system

KU5. use of emission and pollution control devices and measures taken to control pollution

KU6. preventative and remedial actions to be taken in the case of exposure to toxic materials solvents, flux, lead etc.

KU7. precautionary measures taken to prevent fire accident

KU8. various causes of fire (Causes of fires: heating of metal, spontaneous ignition, sparking, electrical heating, loose fires (smoking, welding, etc.), chemical fires, etc.)

KU9. different materials (sand, water, foam, CO₂, dry powder) used in fire extinguisher and proper method of using it to extinguish fire

KU10. various types of safety signs with their names

KU11. basic first aid treatment relevant to the condition e.g. shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries basic techniques of bandaging

KU12. artificial respiration and the CPR Process



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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read & write an accident/ incident report in local language and/or English
- GS2.** read and comprehend basic content to read labels, charts, signages, manuals etc.
- GS3.** question co-workers appropriately and give clear instruction to them and sub ordinates in order to clarify instructions
- GS4.** remain congenial while discussing and debating issues with co-workers
- GS5.** observe appropriate protocols for communication based on situation, hierarchy, organizational culture and practice
- GS6.** plan and organize work schedule, work area, tools, equipment and materials to maintain decorum and for improved productivity
- GS7.** build customer relationships and use customer centric approach
- GS8.** seek assistance from other sources to resolve problem and report problems that cannot be resolved to the appropriate authority

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow workplace health and safety practices</i>	25	36	-	-
PC1. use protective clothing/equipment for specific tasks and work conditions (Protective clothing: leather or asbestos gloves, flame proof aprons, flame proof overalls buttoned to neck, cuff less (without folds) trousers, reinforced footwear, helmets/hard hats, cap and shoulder covers, ear defenders/plugs, safety boots, knee pads, particle masks, glasses/goggles/visors) (Equipment: hand and face shields, machine guards, residual current devices, shields, dust sheets, respirator)	3	5	-	-
PC2. identify job-site hazards and possible causes of risks/accident at the workplace (Hazards: electrical hazards, sharp edged and heavy tools, heated metals, oxyfuel and gas cylinders, welding radiation, hazardous surfaces, hazardous substances, physical hazards. Possible causes of risk and accident: physical actions, not following instructions, inattention, sickness and incapacity, health, not taking safety precautions)	3	2	-	-
PC3. use standard safe working practices when working at heights, confined areas, trenches and with electrical equipment	3	5	-	-
PC4. follow warning signs and symbols while working with electrical systems (Warning signs: Danger plate, high voltage area, out of service, etc.)	2	2	-	-
PC5. ensure positive isolation of electrical equipment and system as per the given standards	2	4	-	-
PC6. identify any abnormalities in electrical equipment or system viz temperature, pressure, flow of current etc.	2	4	-	-
PC7. use various methods of accident prevention in the work environment (Methods of accident prevention: use of equipment and working practices, safety notices, advice, instruction from colleagues and supervisors)	2	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC8. use proper scaffolds, elevated platforms and ladders for working at height	2	3	-	-
PC9. inspect various tools and plants (T & P) routinely for any signs of oil, water leakages	2	3	-	-
PC10. store flammable materials and machine lubricating oil safely and correctly	2	2	-	-
PC11. check that the emission and pollution control devices are working properly in line with environmental policy standards	1	3	-	-
PC12. inform the relevant authorities about any abnormal situation/behavior of any equipment/system without delay	1	3	-	-
<i>Follow fire safety practices</i>	6	12	-	-
PC13. use various appropriate fire extinguishers for different types of fires correctly (Types of fires: Class A, B, C, D and E)	3	5	-	-
PC14. use appropriate rescue techniques during fire hazard	2	5	-	-
PC15. use good housekeeping practices at all times to avoid accidental fire	1	2	-	-
<i>Follow emergencies, rescue and first-aid procedures</i>	8	13	-	-
PC16. administer appropriate first aid to the victims for the relevant injuries (Injuries: Bleeding, burns, choking, electric shock, poisoning etc.)	3	6	-	-
PC17. act promptly and appropriately to an accident situation and administer basic first aid including CPR to the victims of cardiac arrest due to electric shock or medical emergency in real	2	5	-	-
PC18. write an incident report and send the same to the authorized person	3	2	-	-
NOS Total	39	61	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1331
NOS Name	Apply basic health and safety practices for power related work
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	1.5
Version	2.0
Next Review Date	31/08/2026



Qualification Pack

PSS/N1336: Working effectively with others

Description

This unit covers basic practices that improve effectiveness of working with others in an organizational set-up

Scope

The scope covers the following :

- Work effectively with others
- Respect diversity

Elements and Performance Criteria

Work effectively with others

To be competent, the user/individual on the job must be able to:

- PC1.** accept information and instructions from the supervisor and fellow workers and seek clarification when required
- PC2.** pass the information to the persons to all those concerns within agreed timeframe in a manner they can understand and confirm its receipt
- PC3.** help & assist fellow workers in performing their tasks in a positive manner so as to maximize efficiency
- PC4.** use appropriate tone, pitch and language while communicating to convey politeness, assertiveness, care and professionalism
- PC5.** listen attentively and respond appropriately while interacting with others at work
- PC6.** act responsibly and demonstrate disciplined behaviour at the workplace (Disciplined behaviour: Punctuality, completing tasks within agreed timescales, ensuring quality standards, not gossiping and wasting time, honesty, etc.)
- PC7.** escalate grievances and problems to the appropriate authority as per the organizational procedure to resolve them and avoid conflict

Respect diversity

To be competent, the user/individual on the job must be able to:

- PC8.** respect gender diversity in the workplace
- PC9.** use respectful verbal, non-verbal and written communication that is gender, disability and culturally inclusive
- PC10.** communicate with everyone without any personal bias based on gender, disability, caste, religion, colour, sexual orientation and culture
- PC11.** recognize indicators of harassment and discrimination based on gender, disability, caste, religion, colour, sexual orientation and culture at workplace and follow organisational policy for reporting the same

Knowledge and Understanding (KU)



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The individual on the job needs to know and understand:

- KU1.** legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU3.** relevant people and their responsibilities within the work area
- KU4.** escalation matrix and procedures for reporting work and employment related issues
- KU5.** key elements of active listening and assertive communication
- KU6.** importance of teamwork in organizational and individual success
- KU7.** ethics and disciplined behavior for professional success
- KU8.** expressing and addressing grievances appropriately and managing interpersonal conflict effectively
- KU9.** harassment and discrimination based on gender, disability, caste, religion and culture and how to recognize it.
- KU10.** organizational policy/ies for harassment and discrimination

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write clear and legible notes to self, colleagues and seniors to pass messages, keep records, prepare to-do lists, take down instructions
- GS2.** read basic terms and terminologies to accurately interpret work related documents, labels, supervisor instructions in the local language
- GS3.** interact with the supervisor appropriately (correct protocol and manner of speaking) in order to understand the basic requirements of the product, production plans and other associated requirements
- GS4.** display active listening skills while interacting with co-workers and other in the workplace
- GS5.** follow organization's rule-based decision-making process
- GS6.** use appropriate planning to maintain a smooth relationship with fellow team members
- GS7.** meet customer requirements
- GS8.** critically evaluate operation parameters in relation to system normality

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Work effectively with others</i>	18	32	-	-
PC1. accept information and instructions from the supervisor and fellow workers and seek clarification when required	2	4	-	-
PC2. pass the information to the persons to all those concerns within agreed timeframe in a manner they can understand and confirm its receipt	1	4	-	-
PC3. help & assist fellow workers in performing their tasks in a positive manner so as to maximize efficiency	2	4	-	-
PC4. use appropriate tone, pitch and language while communicating to convey politeness, assertiveness, care and professionalism	4	6	-	-
PC5. listen attentively and respond appropriately while interacting with others at work	3	6	-	-
PC6. act responsibly and demonstrate disciplined behaviour at the workplace (Disciplined behaviour: Punctuality, completing tasks within agreed timescales, ensuring quality standards, not gossiping and wasting time, honesty, etc.)	3	5	-	-
PC7. escalate grievances and problems to the appropriate authority as per the organizational procedure to resolve them and avoid conflict	3	3	-	-
<i>Respect diversity</i>	16	24	-	-
PC8. respect gender diversity in the workplace	4	6	-	-
PC9. use respectful verbal, non-verbal and written communication that is gender, disability and culturally inclusive	4	6	-	-
PC10. communicate with everyone without any personal bias based on gender, disability, caste, religion, colour, sexual orientation and culture	4	6	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. recognize indicators of harassment and discrimination based on gender, disability, caste, religion, colour, sexual orientation and culture at workplace and follow organisational policy for reporting the same	4	6	-	-
NOS Total	34	56	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1336
NOS Name	Working effectively with others
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	1
Version	2.0
Last Reviewed Date	22/10/2024
Next Review Date	22/10/2027
NSQC Clearance Date	22/10/2024



Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:



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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e-mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.



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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026



Qualification Pack

PSS/N1717: Prepare testing and maintenance of transformer

Description

This unit is about the installation, testing and maintenance of industrial transformers.

Scope

The scope covers the following :

- Demonstrate procedure of testing and maintaining the transformer

Elements and Performance Criteria

Install the transformer

To be competent, the user/individual on the job must be able to:

- PC1.** identify the terminals of the transformer and find out polarity
- PC2.** trace the primary and secondary winding of the transformer
- PC3.** select, operate and maintain the tools and equipment required for installing and operating a transformer
- PC4.** select and install the transformer as per drawings and documentation provided

Test and maintain the transformer

To be competent, the user/individual on the job must be able to:

- PC5.** connect and test three-phase transformer in a given load
- PC6.**
 - check the transformer for installation resistance (HV
 - LV, LV-E, HV-E)
- PC7.** check the earth resistance
- PC8.** operate multiple transformers by connecting in parallel
- PC9.** perform polarity test on the transformer
- PC10.** test dielectric strength of transformer oil, and its insulation
- PC11.** inspect silica gel, breather, conservator, the temperature of the transformer

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

1. relevant legislation, standards, policies, and procedures followed in the organisation
2. organizational norms relevant to own employment and performance conditions
3. relevant health and safety requirements applicable in the work place
4. own job role and responsibilities and relevant sources for information
5. relevant people and their responsibilities within the work area
6. escalation matrix and procedures for reporting work and employment related issues
7. organizational procedures for requisitioning and indenting materials and equipment



Qualification Pack

8. organisational procedure for recording and reporting job works and related information
9. basic concepts of AC single phase and 3-phase supply
10. conductors, insulators and their types
11. types, uses and parts of a transformer
12. working principle and construction of a transformer and circuit diagram of transformers
13. breakdown voltage test and various protections used for transformers
14. how to test and maintain transformer oil and filtration
15. various preventive and breakdown maintenance work to be performed on transformers
16. concept of the gauge of wire, conductor material and its current carrying capacity and cable binding strap and buttons
17. types of faults and method of fault findings in a transformer
18. construction and the application of auto transformer, CT (current transformer), PT (potential transformer), fuse base and link, connectors, indicator lights, push buttons

Generic Skills (GS)

User/individual on the job needs to know how to:

1. write notes to record information to be used at work like winding data, material requirements, etc.
2. write observations related to the process, if any
3. read dimensions from drawings and diagrams correctly
4. read instructions posted on the equipment/walls/sign boards etc.
5. read safety instructions on machines and devices
6. communicate clearly with team and other staff members
7. discuss task lists, schedules and activities with the team members
8. make appropriate work-related judgments based on laid down policies and procedures
9. plan and organize tasks to meet quantity and quality specifications
10. identify the impact of own work and actions on end customer
11. identify changes in conditions, operations, and the environment to anticipate and address problems



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Install the transformer</i>	10	25	-	-
PC1. identify the terminals of the transformer and find out polarity	2	4	-	-
PC2. trace the primary and secondary winding of the transformer	2	7	-	-
PC3. select, operate and maintain the tools and equipment required for installing and operating a transformer	3	7	-	-
PC4. select and install the transformer as per drawings and documentation provided	3	7	-	-
<i>Test and maintain the transformer</i>	20	45	-	-
PC5. connect and test three-phase transformer in a given load	2	5	-	-
PC6. - check the transformer for installation resistance (HV - LV, LV-E, HV-E)	3	5	-	-
PC7. check the earth resistance	3	5	-	-
PC8. operate multiple transformers by connecting in parallel	3	8	-	-
PC9. perform polarity test on the transformer	3	8	-	-
PC10. test dielectric strength of transformer oil, and its insulation	2	7	-	-
PC11. inspect silica gel, breather, conservator, the temperature of the transformer	4	7	-	-
NOS Total	30	70	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1717
NOS Name	Prepare testing and maintenance of transformer
Sector	Power
Sub-Sector	Generation, Transmission, Distribution
Occupation	Operation and Maintenance - Generation, Erection, Installation and Commissioning - Generation, Erection, Installation and Commissioning - Distribution
NSQF Level	4
Credits	2.0
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQF Clearance Date	31/08/2023



Qualification Pack

PSS/N1716: Prepare control panel wiring of industrial AC

Description

This unit is about performing the control panel wiring of industrial AC motors and the preparation required.

Scope

The scope covers the following :

- Perform preparatory activities for control panel wiring of industrial AC Motors.

Elements and Performance Criteria

Prepare for control panel wiring of industrial AC Motors

To be competent, the user/individual on the job must be able to:

- PC1.** perform various wire bending operations
- PC2.** perform stripping and glanding of wires and cables
- PC3.** perform crimping on thimbles, lugs
- PC4.** build contactor control circuits -push button operation, logic building using nc/no (normally open/normally closed) controls
- PC5.** assemble various wiring accessories
- PC6.** select, operate and maintain the tools and equipment required for installing and operating an ac motor

Perform control panel wiring of industrial AC Motors

To be competent, the user/individual on the job must be able to:

- PC7.** connect ac motor as per instructions provided including structured cabling systems as per manufacturer's instructions
- PC8.** assemble and replace bearings with proper equipment as per standards
- PC9.** set-up equipment to control the speed and torque of an AC motor
- PC10.** control three phase induction motor using
- PC11.** troubleshoot and take general care maintenance of AC motor

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

1. relevant legislation, standards, policies, and procedures followed in the organisation
2. organizational norms relevant to own employment and performance conditions
3. relevant health and safety requirements applicable in the work place
4. own job role and responsibilities and relevant sources for information
5. relevant people and their responsibilities within the work area
6. escalation matrix and procedures for reporting work and employment related issues
7. organizational procedures for requisitioning and indenting materials and equipment



Qualification Pack

8. organisational procedure for recording and reporting job works and related information
9. basic concepts regarding ac single phase and 3-phase supply
10. measurement of electrical quantities such as current, voltage, resistance, impedance, power factor and energy
11. importance of basic laws ohms law and kirchhoffs law
12. the working, construction and circuit diagram of A.C Motor
13. types of single phase and three phase motors and their working, slip ring and squirrel cage induction motor
14. how to control the speed and torque of an AC motor
15. AC drive and its connection, configuration and parameterization
16. the symbols of electrical parts
17. control and power circuits, contactors, overload relays, molded case circuit breakers (MCCb), motor protection relays, motor protection circuit breakers etc.
18. different types of standards, drawings, installation descriptions and manuals
19. functioning and usage of measuring tools, wire gauges and tools required for marking, punching, cutting, drilling, filing, stripping, crimping, socketing and fixing glands and screws etc.
20. conductors and their joints, the concept of gauge of wire, conductor material and its current carrying capacity
21. types of faults and method of fault findings
22. cable binding strap and buttons, crimping and crimping tools, soldering
23. CT (current transformer), PT (potential transformer), fuse base and link, connectors, indicator lights, push buttons

Generic Skills (GS)

User/individual on the job needs to know how to:

1. write notes to record information to be used at work like winding data, material requirements, etc.
2. write observations related to the process, if any
3. read dimensions from drawings and diagrams correctly
4. read instructions posted on the equipment/walls/ sign boards etc.
5. read safety instructions on machines and devices
6. communicate clearly with team and other staff members
7. discuss task lists, schedules and activities with the team members
8. make appropriate work-related judgments based on laid down policies and procedures
9. plan and organize tasks to meet quantity and quality specifications
10. identify the impact of own work and actions on end customer
11. identify changes in conditions, operations, and the environment to anticipate and address problems



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for control panel wiring of industrial AC Motors</i>	17	39	-	-
PC1. perform various wire bending operations	3	7	-	-
PC2. perform stripping and glanding of wires and cables	3	7	-	-
PC3. perform crimping on thimbles, lugs	3	6	-	-
PC4. build contactor control circuits -push button operation, logic building using nc/no (normally open/normally closed) controls	3	6	-	-
PC5. assemble various wiring accessories	2	7	-	-
PC6. select, operate and maintain the tools and equipment required for installing and operating an ac motor	3	6	-	-
<i>Perform control panel wiring of industrial AC Motors</i>	13	31	-	-
PC7. connect ac motor as per instructions provided including structured cabling systems as per manufacturer's instructions	2	5	-	-
PC8. assemble and replace bearings with proper equipment as per standards	3	7	-	-
PC9. set-up equipment to control the speed and torque of an AC motor	3	7	-	-
PC10. control three phase induction motor using	2	6	-	-
PC11. troubleshoot and take general care maintenance of AC motor	3	6	-	-
NOS Total	30	70	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1716
NOS Name	Prepare control panel wiring of industrial AC
Sector	Power
Sub-Sector	Generation, Transmission, Distribution
Occupation	Operation and Maintenance - Generation, Erection, Installation and Commissioning - Generation, Erection, Installation and Commissioning - Distribution
NSQF Level	4
Credits	2.0
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023



Qualification Pack

PSS/N1715: Prepare industrial wiring as per planned

Description

This unit is about wiring up various types of industrial wiring and the preparation required.

Scope

The scope covers the following :

- Perform preparatory activities for industrial wiring operations

Elements and Performance Criteria

Prepare for industrial wiring operations

To be competent, the user/individual on the job must be able to:

- PC1.** measure current, voltage and resistance of single phase and three phase load
- PC2.** make simple single phase and three phase electrical connections using resistance, voltmeter, ammeter and multimeter etc.
- PC3.** verify the characteristics of series, parallel and its combination circuits
- PC4.** identify the phase, neutral and earth connections in single and three phase supply
- PC5.**
 - read and interpret drawings and documentation including 1 system through rlc circuits with phase diagram, 3 system
 - generation of 3 system, 3 type (star, delta)
- PC6.** identify, use and maintain various hand tools and measuring instruments
- PC7.** plan ducting and wiring systems for industrial use and when and where to use a specific ducting and/or wiring system
- PC8.** plan staircase wiring and master control wiring using drawings and documentation provided
- PC9.** demonstrate the usage of direct on-line starter and star-delta starter

Perform different types of industrial wiring

To be competent, the user/individual on the job must be able to:

- PC10.** install various electrical power supplies, such as single phase, three-phase, direct current and low voltage
- PC11.** install structured cabling systems such as computer network cabling, fire/burglar alarm), control and monitoring, access control closed circuit television
- PC12.** assemble different termination adaptors, including glands onto duct and attach ducts, of different types, securely onto a surface
- PC13.** install different types of installations for illumination: fluorescent light, metal halide, high pressure mercury vapor, high pressure sodium vapor, led, flame proof lighting fixtures
- PC14.** test the industrial illumination wiring installations for insulation resistance using a megger
- PC15.** perform troubleshooting of faults present in the industrial wiring

Knowledge and Understanding (KU)



Qualification Pack

The individual on the job needs to know and understand:

1. relevant legislation, standards, policies, and procedures followed in the organisation
2. organizational norms relevant to own employment and performance conditions
3. relevant health and safety requirements applicable in the work place
4. job role and responsibilities and relevant sources for information
5. about the people and their responsibilities within the work area
6. escalation matrix and procedures for reporting work and employment related issues
7. organizational procedures for requisitioning and indenting materials and equipment
8. organisational procedure for recording and reporting job works and related information
9. basic concepts regarding ac single phase and 3-phase supply
10. measurement of electrical quantities such as current, voltage, resistance, impedance, power factor and energy
11. importance of basic laws ohms law and kirchhoffs law
12. different types of electrical wiring such as residential, industrial and overhead lines
13. correct operation of the electrical installation in accordance with the planned specification
14. conversion of star to delta and vice versa
15. cea regulation for measures relating to safety and electric supply (2010)
16. layout and drawings of the circuit
17. types of electric lighting and heating systems for commercial, domestic residential and industrial use
18. how to plan installation work using drawings and documentation provided
19. different types of standards, drawings, installation descriptions and manuals
20. functioning and usage of tools required for marking, punching, cutting, drilling, filing, stripping, crimping, socketing and fixing glands and screws etc.
21. functioning and usage of measuring tools, wire gauges etc.
22. classification and identification of the electrical equipment, cables, wires and electrical accessories used in industry
23. how to lay out cable and in different circuits
24. conductors, the concept of gauge of wire, conductor material and its current carrying capacity
25. determination of fuse size according to the load of the circuit and its location
26. how to use megger and test lamps in fault location
27. types of faults and method of fault findings
28. different types of switchgears used in general electrical installations
29. cable binding strap and buttons, crimping and crimping tools, soldering
30. CT (current transformer), pt (potential transformer), fuse base and link, connectors, indicator lights, push buttons

Generic Skills (GS)

User/individual on the job needs to know how to:



Qualification Pack

1. write notes to record information to be used at work like winding data, material requirements, etc.
2. write observations related to the process, if any
3. read dimensions from drawings and diagrams correctly
4. read instructions posted on the equipment/walls/sign boards etc.
5. read safety instructions on machines and devices
6. communicate clearly with team and other staff members
7. discuss task lists, schedules and activities with the team members
8. make appropriate work-related judgments based on laid down policies and procedures
9. plan and organize tasks to meet quantity and quality specifications
10. identify impact of own work and actions on end customer
11. identify changes in conditions, operations, and the environment to anticipate and address problems

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for industrial wiring operations</i>	17	42	-	-
PC1. measure current, voltage and resistance of single phase and three phase load	2	4	-	-
PC2. make simple single phase and three phase electrical connections using resistance, voltmeter, ammeter and multimeter etc.	1	4	-	-
PC3. verify the characteristics of series, parallel and its combination circuits	2	4	-	-
PC4. identify the phase, neutral and earth connections in single and three phase supply	2	3	-	-
PC5. • read and interpret drawings and documentation including 1 system through rlc circuits with phase diagram, 3 system • generation of 3 system, 3 type (star, delta)	2	7	-	-
PC6. identify, use and maintain various hand tools and measuring instruments	1	4	-	-
PC7. plan ducting and wiring systems for industrial use and when and where to use a specific ducting and/or wiring system	3	7	-	-
PC8. plan staircase wiring and master control wiring using drawings and documentation provided	2	6	-	-
PC9. demonstrate the usage of direct on-line starter and star-delta starter	2	3	-	-
<i>Perform different types of industrial wiring</i>	13	28	-	-
PC10. install various electrical power supplies, such as single phase, three-phase, direct current and low voltage	2	6	-	-
PC11. install structured cabling systems such as computer network cabling, fire/burglar alarm), control and monitoring, access control closed circuit television	2	6	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. assemble different termination adaptors, including glands onto duct and attach ducts, of different types, securely onto a surface	2	4	-	-
PC13. install different types of installations for illuminationillumination: fluorescent light, metal halide, high pressure mercury vapor, high pressure sodium vapor, led, flame proof lighting fixtures	3	5	-	-
PC14. test the industrial illumination wiring installations for insulation resistance using a megger	2	3	-	-
PC15. perform troubleshooting of faults present in the industrial wiring	2	4	-	-
NOS Total	30	70	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1715
NOS Name	Prepare industrial wiring as per planned
Sector	Power
Sub-Sector	Generation, Transmission, Distribution
Occupation	Operation and Maintenance - Generation, Erection, Installation and Commissioning - Generation, Erection, Installation and Commissioning - Distribution Downstream
NSQF Level	4
Credits	2.0
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023



Qualification Pack

PSS/N1714: Preparation installation and testing of Switchgears

Description

This unit is about installation, testing and maintenance of switchgears and cables.

Scope

The scope covers the following :

- Demonstrate procedure of installation and testing of various cables and cable joints.

Elements and Performance Criteria

Perform installation and testing of switchgears

To be competent, the user/individual on the job must be able to:

- PC1.** assemble switchboard apparatus as per layout drawings
- PC2.** install electrical switchboards onto a surface as per drawings and documentation provided
- PC3.** test installations before energizing and check for proper connection
- PC4.** test installations when energized by checking complete function of all equipment to ensure the correct operation of new installations as per instructions

Perform installation and testing of various cables and cable joints

To be competent, the user/individual on the job must be able to:

- PC5.** identify sub-circuits and cables for connection of security control panel
- PC6.** install and terminate cables to connect the security control panel to existing sub-circuit
- PC7.** perform different types of cable jointing
- PC8.** perform different types of cable trays bending in the range of 45 to 90 degrees
- PC9.** perform preventive maintenance of various switchboard apparatus
- PC10.** inspect, test, rectify abnormal conditions, and commission connection of the circuit
- PC11.** troubleshoot electrical installations and identify faults in the cable

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

1. relevant legislation, standards, policies, and procedures followed in the organisation
2. organizational norms relevant to own employment and performance conditions
3. relevant health and safety requirements applicable in the work place
4. own job role and responsibilities and relevant sources for information
5. relevant people and their responsibilities within the work area
6. escalation matrix and procedures for reporting work and employment related issues
7. organizational procedures for requisitioning and indenting materials and equipment
8. organisational procedure for recording and reporting job works and related information



Qualification Pack

9. basic concepts regarding ac single phase and 3-phase supply
10. measurement of electrical quantities such as current, voltage, resistance, impedance, power factor and energy
11. importance of basic laws - ohms law and kirchhoffs law
12. correct operation of the electrical installation in accordance with the planned specification
13. types of switches such as SPST, SPDT, DPST, DPDT, Toggle switch, push button switch, level actuator, limit switch, selector switch, flame proof switch
14. definition, use, types and applications of circuit breakers such as MCB (Miniature Circuit Breaker), MCCB (Molded Case Circuit Breakers), ELCB (Earth-leakage Circuit Breaker), ACB (air circuit breaker), SF6 circuit breaker, VCB (Vacuum Circuit Breaker), flame proof switchgear specification
15. types and sizes of cables, specification current rating and their application
16. how to lay out cable and in different circuits
17. different types of joints in the cable such as straight through, long routes and repaired section, branch y joints for branch of a sector, t- joint for branching of a sector, transition joints special joint between two different types of cable
18. how to plan installation work using drawings and documentation provided
19. classification and identification of the electrical equipment, cables, wires and electrical accessories used in industry, conductors, insulators, their types and their joints, concept of gauge of wire, conductor material and its current carrying capacity
20. determination of fuse size, MCB, MCCB etc. according to the load of circuits and its location
21. cable binding strap and buttons
22. CT (current transformer), PT (potential transformer), fuse base and link, connectors, indicator lights, push buttons

Generic Skills (GS)

User/individual on the job needs to know how to:

1. write notes to record information to be used at work like winding data, material requirements, etc.
2. write observations related to the process, if any
3. read dimensions from drawings and diagrams correctly
4. read instructions posted on the equipment/walls/sign boards etc.
5. read safety instructions on machines and devices
6. communicate clearly with team and other staff members
7. discuss task lists, schedules and activities with the team members
8. make appropriate work related judgements based on laid down policies and procedures
9. plan and organize tasks to meet quantity and quality specifications
10. identify impact of own work and actions on end customer
11. identify changes in conditions, operations, and the environment to anticipate and address problems



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform installation and testing of switchgears</i>	9	28	-	-
PC1. assemble switchboard apparatus as per layout drawings	2	7	-	-
PC2. install electrical switchboards onto a surface as per drawings and documentation provided	3	8	-	-
PC3. test installations before energizing and check for proper connection	2	6	-	-
PC4. test installations when energized by checking complete function of all equipment to ensure the correct operation of new installations as per instructions	2	7	-	-
<i>Perform installation and testing of various cables and cable joints</i>	21	42	-	-
PC5. identify sub-circuits and cables for connection of security control panel	2	3	-	-
PC6. install and terminate cables to connect the security control panel to existing sub-circuit	5	7	-	-
PC7. perform different types of cable jointing	3	7	-	-
PC8. perform different types of cable trays bending in the range of 45 to 90 degrees	4	6	-	-
PC9. perform preventive maintenance of various switchboard apparatus	2	7	-	-
PC10. inspect, test, rectify abnormal conditions, and commission connection of the circuit	3	6	-	-
PC11. troubleshoot electrical installations and identify faults in the cable	2	6	-	-
NOS Total	30	70	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1714
NOS Name	Preparation installation and testing of Switchgears
Sector	Power
Sub-Sector	Generation, Transmission, Distribution
Occupation	Operation and Maintenance - Generation, Erection, Installation and Commissioning - Generation, Erection, Installation and Commissioning - Distribution
NSQF Level	4
Credits	2.0
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023



Qualification Pack

PSS/N1713: Preparation pipe and plate earthing

Description

This unit is about planning, preparing and installation of pipe and plate earthing as per standard procedures.

Scope

The scope covers the following :

- Perform installation of plate and pipe earthing

Elements and Performance Criteria

Plan and prepare for pipe and plate earthing

To be competent, the user/individual on the job must be able to:

- PC1.** Identify and obtain specific tools, materials and equipment required for earthing
- PC2.** Ensure that the earth pit excavation is done as per specifications
- PC3.** Carry out testing and maintenance of Earth Resistance

Install plate earthing

To be competent, the user/individual on the job must be able to:

- PC4.** Insert the galvanized iron plate electrode in case of plate earthing
- PC5.** Fix the pipe electrode as per specifications in case of pipe earthing

Install pipe earthing

To be competent, the user/individual on the job must be able to:

- PC6.** Check that the connections done are not loose and fixed as per specifications
- PC7.** Install water connection pipeline as per requirement in case of plate earthing

Perform post earthing procedures

To be competent, the user/individual on the job must be able to:

- PC8.** Fill the annular space between the electrode and borehole walls with alternating layers of coke or charcoal and common salt in case of pipe earthing
- PC9.** Follow the approved design and drawing for fixing and laying of earth wires or copper strips between the earth electrode and the electrical room

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

1. Relevant legislation, standards, policies, and procedures followed in the organisation
2. Organizational norms relevant to own employment and performance conditions
3. Relevant health and safety requirements applicable in the workplace
4. Own job role and responsibilities and relevant sources for information
5. Relevant people and their responsibilities within the work area



Qualification Pack

6. Escalation matrix and procedures for reporting work and employment related issues
7. Organizational procedures for requisitioning and indenting materials and equipment
8. Organisational procedure for recording and reporting job works and related information
9. Basic concepts of AC single phase and 3-phase supply
10. Conductors, insulators and their types
11. Concept of earthing, purpose and types
12. How to measure earth resistance
13. Basics of masonry and construction
14. The requirement to fill the annular space between the earth electrode and borehole walls and permissible materials which can be used

Generic Skills (GS)

User/individual on the job needs to know how to:

1. Write notes to record information to be used at work like material requirements, etc.
2. Write observations related to the process, if any
3. Read dimensions from drawings and diagrams correctly
4. Read instructions posted on the equipment/walls/ sign boards etc.
5. Read safety instructions on machines and devices
6. Communicate clearly with team and other staff members
7. Discuss task lists, schedules and activities with the team members
8. Make appropriate work-related judgments based on laid down policies and procedures
9. Plan and organize tasks to meet quantity and quality specifications
10. Identify the impact of own work and actions
11. identify changes in conditions, operations, and the environment to anticipate and address



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Plan and prepare for pipe and plate earthing</i>	10	19	-	-
PC1. Identify and obtain specific tools, materials and equipment required for earthing	3	5	-	-
PC2. Ensure that the earth pit excavation is done as per specifications	3	5	-	-
PC3. Carry out testing and maintenance of Earth Resistance	4	9	-	-
<i>Install plate earthing</i>	7	17	-	-
PC4. Insert the galvanized iron plate electrode in case of plate earthing	4	8	-	-
PC5. Fix the pipe electrode as per specifications in case of pipe earthing	3	9	-	-
<i>Install pipe earthing</i>	5	18	-	-
PC6. Check that the connections done are not loose and fixed as per specifications	2	8	-	-
PC7. Install water connection pipeline as per requirement in case of plate earthing	3	10	-	-
<i>Perform post earthing procedures</i>	8	16	-	-
PC8. Fill the annular space between the electrode and borehole walls with alternating layers of coke or charcoal and common salt in case of pipe earthing	4	8	-	-
PC9. Follow the approved design and drawing for fixing and laying of earth wires or copper strips between the earth electrode and the electrical room	4	8	-	-
NOS Total	30	70	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1713
NOS Name	Preparation pipe and plate earthing
Sector	Power
Sub-Sector	Generation, Transmission, Distribution
Occupation	Operation and Maintenance - Generation, Erection, Installation and Commissioning - Distribution, Erection, Installation and Commissioning - Generation
NSQF Level	4
Credits	3.0
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQF Clearance Date	31/08/2023



Qualification Pack

PSS/N1712: Preparation for the operation of DG (Diesel Generator) sets and its controls

Description

This unit is about the preparation and operation of DG (Diesel Generator) set and its control.

Scope

The scope covers the following :

- Demonstrate preparatory activities for the operation of DG sets

Elements and Performance Criteria

Preparation for the operation of DG (Diesel Generator) Sets

To be competent, the user/individual on the job must be able to:

- PC1.** Check the conditions for starting the DG (Diesel Generator) set
- PC2.** Change lubricant, coolant etc., if required
- PC3.** Select, operate and maintain the tools and equipment required for installing and operating a DG set

Operate DG (Diesel Generator) set and its controls

To be competent, the user/individual on the job must be able to:

- PC4.** Install the required DG set as per drawings and documentation provided
- PC5.** Start, run and load the DG set
- PC6.** Conduct a test process to ensure the performance of installed electrical equipment is as per the defined specifications
- PC7.** Identify the parts of the lead acid battery and test for its condition
- PC8.** Check the quantity of electrolyte
- PC9.** Perform charging of lead acid battery and test it
- PC10.** Perform troubleshooting and general care and maintenance of the DG set

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

1. relevant legislation, standards, policies, and procedures followed in the organisation
2. organizational norms relevant to own employment and performance conditions
3. relevant health and safety requirements applicable in the work place
4. own job role and responsibilities and relevant sources for information
5. relevant people and their responsibilities within the work area
6. escalation matrix and procedures for reporting work and employment related issues
7. organizational procedures for requisitioning and indenting materials and equipment



Qualification Pack

8. organisational procedure for recording and reporting job works and related information
9. basic concepts of AC single phase and 3-phase supply
10. conductors, insulators, their types and cable binding strap and buttons
11. construction and working of a generator and the DG control panel
12. characteristics and circuit diagram of a generator
13. concept of the gauge of wire, conductor material and its current carrying capacity
14. concept of different types of switchgears used in general electrical installations
15. control and power circuits, contactors, overload relays, moulded case circuits breakers, motor protection relays, motor protection circuit breakers etc.
16. CT (current transformer), PT (potential transformer), fuse base and link, connectors, indicator lights, push buttons
17. the functioning and maintenance procedures of various parts of a lead acid battery
18. the functioning and preventive maintenance procedures of various parts of dg sets and routine cleaning of filters etc.

Generic Skills (GS)

User/individual on the job needs to know how to:

1. write notes to record information to be used at work like material requirements, etc.
2. write observations related to the process, if any
3. read dimensions from drawings and diagrams correctly
4. read instructions posted on the equipment/walls/sign boards etc.
5. read safety instructions on machines and devices
6. communicate clearly with team and other staff members
7. discuss task lists, schedules and activities with the team members
8. make appropriate work-related judgments based on laid down policies and procedures
9. plan and organize tasks to meet quantity and quality specifications
10. identify the impact of own work and actions on end customer
11. identify changes in conditions, operations, and the environment to anticipate and address problems

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Preparation for the operation of DG (Diesel Generator) Sets</i>	9	20	-	-
PC1. Check the conditions for starting the DG (Diesel Generator) set	2	5	-	-
PC2. Change lubricant, coolant etc., if required	3	7	-	-
PC3. Select, operate and maintain the tools and equipment required for installing and operating a DG set	4	8	-	-
<i>Operate DG (Diesel Generator) set and its controls</i>	21	50	-	-
PC4. Install the required DG set as per drawings and documentation provided	4	8	-	-
PC5. Start, run and load the DG set	3	7	-	-
PC6. Conduct a test process to ensure the performance of installed electrical equipment is as per the defined specifications	3	8	-	-
PC7. Identify the parts of the lead acid battery and test for its condition	1	5	-	-
PC8. Check the quantity of electrolyte	2	6	-	-
PC9. Perform charging of lead acid battery and test it	4	8	-	-
PC10. Perform troubleshooting and general care and maintenance of the DG set	4	8	-	-
NOS Total	30	70	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1712
NOS Name	Preparation for the operation of DG (Diesel Generator) sets and its controls
Sector	Power
Sub-Sector	Generation, Transmission, Distribution
Occupation	Operation and Maintenance - Generation, Erection, Installation and Commissioning - Generation
NSQF Level	4
Credits	2.0
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass 70% aggregate for the QP.



Qualification Pack

7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
PSS/N1331.Apply basic health and safety practices for power related work	39	61	0	0	100	5
PSS/N1336.Working effectively with others	34	56	0	0	90	5
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
PSS/N1717.Prepare testing and maintenance of transformer	30	70	0	0	100	10
PSS/N1716.Prepare control panel wiring of industrial AC	30	70	0	0	100	15
PSS/N1715.Prepare industrial wiring as per planned	30	70	0	0	100	15
PSS/N1714.Preparation installation and testing of Switchgears	30	70	0	0	100	10
PSS/N1713.Preparation pipe and plate earthing	30	70	0	0	100	15
PSS/N1712.Preparation for the operation of DG (Diesel Generator) sets and its controls	30	70	0	0	100	15
Total	273	567	0	0	840	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.